

PRODUCTION AND INVESTIGATION OF PHYSICAL AND TENSILE PROPERTIES OF WASTE POLYETHYLENE – BAOBAB FRUIT SHELL POWDER COMPOSITE

¹Mohammed Sani Yunusa, ¹Ibrahim Abdullahi, ²Nuhu Lawal, and ³Muhammad Tanko Baba

¹Department of Mechanical Engineering, Bayero University Kano, Nigeria.

²Department of Polymer and Textile engineering, Ahmadu Bello University Zaria, Nigeria.

³Department of Mechanical Engineering, Federal Polytechnic Mubi, Adamawa State, Nigeria.

Corresponding author email: muhammadsani1889@gmail.com

ABSTRACT

In this study, the physical and tensile properties of the Waste polyethylene-baobab fruit shell composite were investigated. The highest and the lowest water absorption capacity of the samples were observed to be 12.24 % at 30 wt% and 0.02 % at 0 wt% respectively. the lowest density value (930 kgm^{-3}) at 0 wt% BFS and the highest value (1110 kgm^{-3}) at 30 wt% BFS. The tensile properties showed lowest value of Tensile strength of 10.722 MPa at 0 wt% BFS and increased to highest value (13.827) MPa at 15 wt% BFS, Tensile modulus from 0.871 GPa at 15 wt% BFS to highest value of 1.448 GPa at 30 wt% BFS. Elongation at break were 15.153% at 15wt% BFS and 8.404% 30wt% being the lowest and largest values respectively. The morphology and dispersion of the composites revealed poor filler dispersion.

Keywords: Tensile properties, waste polyethylene, baobab fruit shell powder and composite

1.0 INTRODUCTION

In the world today the need for the development of new products with more efficient material is very important. This makes composites play a major role considering their composition (Santhosh *et al.*, 2014). They consist of reinforcement embedded in a matrix, mixed and bonded together on a macroscopic scale (Shubhra and Alam 2013). This excellent combination gives improved properties which cannot be obtained with any single material (Mallik *et al.*, 2015). Composites made from polymeric materials are gaining dominance as important substitute for metals in various industrial applications such as automotive, marine, sporting goods, aerospace, electronics and construction industries due to their significant properties such as light weight and corrosion resistance (Almaadeed *et al.*, 2013). Particulate reinforced composites have high thermal conductivity and are isotropic in nature. Fillers in particulate composites can be glass, carbon, aramid etc (Vijay *et al.*, 2017).

The distribution of Baobab Trees (*Adansonia digitata* L.) across Southern Africa is estimated to be about 28 million trees. This gives an annual yield of 673,000 tons of whole baobab fruit and

108,000 tons of baobab fruit pulp powder. An average mature baobab fruit produces about 200 kg of fruit per season. The fruit shell of baobab tree hangs singly on long stalks with ovoid, woody and indehiscent shell 20 to 30cm long and up to 10cm in diameter. The shell contains numerous hard brownish seeds, round or ovoid up to 15mm long which are embedded in a yellowish-white floury acidic pulp. The ripe fruit pulp appeared as naturally dehydrated, powdery whitish colored with slightly acidulous taste mostly use as food and drinks. There is presently less economic importance attached to the shells, they are often discarded abundantly, thereby littering the environment (Kmatou, 2011).

Musa *et al.*, (2018) studied the mechanical properties of a composite produced from Recycled Low Density Polyethylene (RLDPE) with sawdust powder filler. The study reported an increase in hardness with increase in filler loading, while the impact strength and compression strength decrease with filler loading. Ashish *et al.*, (2022) investigated the properties of composite developed from silica sand and recycled thermoplastics. The maximum compressive and flexural strength were found to be 46.20 N/mm^2 and

6.24N/mm² with minimum water absorption and sliding wear rate of 0.039% and 0.413×10^{-8} kg/m respectively. The hardness of a composite developed from High density Polyethylene and low Density polyethylene and sand particles at %0wt% LPE, 20wt%HDPE and 30wt% sand particles is 37.0 HV. Negasi *et al*, (2020) investigated the properties of developed recycled Low Density Polyethylene reinforced with Pineapple Leaf Fiber (PALF) composite. The result showed an increase in tensile, impact and flexural strength to the optimal level of 30% fiber weight. The water absorption increased with increase PALF weight. Alexander *et al*, (2022) investigated the mechanical properties of Recycled Low Density Polyethylene (RLDPE) filled with sawdust powder. The study reported an increase in hardness with increase in filler loading, while the impact strength and compression strength decrease with filler loading. Pankaj (2022) conducted a study on development of sand-plastic composites using silica sand and recycled thermoplastics mixture from post-consumer products. The developed composite was characterized for water absorption compressive strength, flexural strength and sliding wear. The maximum compressive and flexural strength obtained were 46.20 N/mm² and 6.24 N/mm² respectively while the minimum water absorption (%) and wear rate were found to be 0.039 and 0.143×10^{-8} kg/m respectively. Athanas (2011) investigate the use of plastic waste as binding material in the development of composite using sand particles as reinforcement. It was found composite produced using plastic waste has porosity value less than 5% while that of micro-concrete is more than 10% and with more abrasion resistance when compared to other composite. Omosebi and Abbas, (2021) conducted a study on a composite produced from recycled pet bottles wastes with improved strength and reduced flammability. The highest compressive strength was at 30wt% and 70wt% respectively is 6.88MPa.

Polyethylene is commonly used in food packaging. It is a popular synthetic polymer with huge hydrophobicity and molecular weight. Since polyethylene is not biodegradable, using it for the production of disposal or packing materials causes many problems, the main problem being

environmental pollution and land shortage problems for landfills (Temitope *et al.*, 2022).

Findings from this work indicate that valorisation of LDPE waste sachet baobab fruit shells can be a value-addition, affordable waste management method for producing biocomposites towards building applications. The baobab tree, also known as the tree of life is a common tree found in northern Nigeria, and widely used as food, medicine and shelter. Its fruit shells are waste products of many local industries. In some part of northern Nigeria such as Borno State baobab fruit shells are discarded or used as fodder for domestic farm animals. In Nigeria, baobab fruit shells are abundant in the Northern region and serve as a source of income for farmers. Despite the abundance of fruit shell, their use as fillers in composites has not been widely published. Therefore, fruit shells as bio-filler can add value to it.

2.0 MATERIALS AND METHODS

2.1 Materials

Baobab fruit shell and waste polyethylene weighted about 5.0 and 7.0 kg respectively, were sourced from Mayo-Belwa local government area, Adamawa state, Nigeria.

2.2 Methods

The Baobab fruit shell and waste polyethylene were manually washed using tap water to remove dust, dirt and sand particles, air dried. The baobab fruit shell was ground into powder using grinding machine and sieved to 250µm using standard test sieve to obtain finer and uniform particle size while the polyethylene waste broken into smaller pieces using knife.

Fabrication of the composites was carried out by compounding and compression moulding techniques at 20 % filler loading using two rolls mill and compression moulding machines. A steel mould of 150 × 120 × 5 mm was used. The control sample was the unfilled waste polypropylene. The matrix and filler were mixed for 5 min in two rolls mill to achieve a homogenous mixture. The mixture was placed in the mould treated with releasing agent. The maximum pressing temperature, pressure, time and cold pressing or pressure holding time were 160 °C, 5 N/mm², 15 min and 5 min respectively (Lawal *et al.*, 2023). After cooling, the composite was removed and conditioned in an oven at 70 °C for 48 hr before testing.

Table 1. Formulation of wPE / BFS Composites

Sample ID	A	B	C	D	E	F	G
wPE wt%	100	95	90	85	80	75	70
BFS wt%	0	5	10	15	20	25	30

2.3 Physical Properties

2.3.1 Water Absorption Test

The test was conducted in accordance with ASTM D570-98, in which the specimens were immersed in water for 24 h at a temperature $23 \pm 1^\circ\text{C}$, then for another 24 h until no further weight gain is noticed. The weight gain was then measured using Sartorius MC201P high sensitivity balance 20 min after being removed from the water, for each immersion. The mass of water absorbed was calculated from the following equation:

$$M_{wabs} = M_{avf} - M_{avi} \dots \dots \dots 2.1$$

Where: M_{wabs} is the mass of water absorbed

M_{avf} is the average final mass after immersion

M_{avi} is the average initial mass before immersion

The percentage water absorbed is given by the equation:

$$\%M_{wabs} = \frac{(M_{avf} - M_{avi})}{M_{avi}} \dots \dots \dots 2.2$$

2.3.2 Density Determination

The density of the composite was determined in accordance with ASTM standard D 792-13 (Method A) by using a Sartorius MC201P high sensitivity balance. Before the Measurements, all samples were dried at 50°C for 18 hours. The specimens were weighed in air using the balance with a precision of 0.1 mg after which were weighed when immersed in distilled water at 21°C using a sinker and wire to hold the specimen completely submerged as required. Precautions were taken to ensure that no air bubbles were trapped prior to taking the reading. The averages of samples taken were calculated as

composite density. The following equations were used to identify the specific gravity and density of the BFS/ rPE composites.

$$Sg^{21^\circ\text{C}} = a/(a + W - b) \dots \dots \dots 2.3$$

Where: $Sg^{21^\circ\text{C}}$ is specific gravity of the composites at 21°C ; a is the mass of specimen, without sinker, in air; b is the mass of specimen and sinker completely immersed and of the wire partially immersed in gas-free distilled water at 21°C ; W is the mass of totally immersed sinker and partially immersed wire.

$$D^{21^\circ\text{C}} = Sg^{21^\circ\text{C}} \times 0.99799 \dots \dots \dots 2.4$$

Where: $D^{21^\circ\text{C}}$ is the density of the composites at 21°C ; 0.99799 g/cm³ is density of the water at 21°C .

2.4 Tensile Properties

2.4.1 Tensile Test

Tensile strength is a measurement of the ability of a material to withstand forces that tend to pull or stretch it apart before breaking. The aim of the experiment was to determine the values of tensile modulus, percentage elongation and maximum tensile strength of the composite material. The test was conducted as per ASTM D3039 (ASTM 2014) using specimen of dumbbell shape with a width of 10mm, length of 100mm and thickness of 3mm. Two fine marks were made on the samples with distance of 40mm apart as the gauge length. The sample was mounted on the tensometer. It was ensured that the ends of the test piece were fitted into the grip of the tensometer. The grip was held between the tension head and operating screw. The load was applied gradually by turning the operating handle. The force applied to the specimen was transmitted to the chart roller which is attached

to a graph paper. Turning of the operating handle leads to gradual increase of the load at the middle span until fracture occur. The test was carried out at turning speed of 50 mm/min. The force and extension was obtained from the load and extension curve from the graph paper. Two specimens from each composition of were used.

2.4.2 Morphological Properties

To study the morphological properties of baobab fruit shell powder / waste polyethylene composites, tensile fractured specimens were viewed using scanning electron microscope (SEM), PHENOM PRO X (Phenomworld, Eindhoven, Netherlands) (ASTM E986-04 2017). The sample with appropriate size was fitted in the specimen compartment, mounted and clipped on a holder. Before observation, the sample was made to be electrically conductive by coating it with platinum. This was achieved with a gold–palladium alloy in an argon-inert environment.

3.0 RESULTS AND DISCUSSION

3.1 Water Absorption

The effects of filler loading on water absorption property for recycled Polyethylene-Baobab shell composites for 504hrs immersion are showed in Figure 4.1. The absorption of water is an important parameter while working with cellulose-based fillers, because high humidity worsens the mechanical properties of the material and change the dimensions. The effect of water

absorption is an important consideration where the material has been designed for applications in contact with water. From the results the ability of the composites to absorb water after exposure at ambient temperature was studied during (504Hours) twenty-one days. The results revealed that percentage (%) water absorption of recycled Polyethylene-Baobab shell composites increased with increase in BFS filler content. This increase is attributed to the water ability to saturate the surface of the material and diffused into the core of the material with the help of non-fully embedded hydrophilic filler. The highest water absorbed by the composite is 12.24%. The maximum water absorption of floor tile composite developed from recycled PET reinforced with natural fiber at 30wt% and 70wt% respectively is 0.98% (Omosebi and Abbas, 2021). Figure 4.1 shows the amount of water absorbed by the composites increased with increase in the Baobab shell powder content in the composites. This is in agreement with results of similar works by Danladi and Tunde (2013). This property is due an increase in the amount of hydrophilic filler in the composites. This is attributed to hydrophilic nature of the Baobab shell particles (BSP) capable of absorbing water, this is to say that the higher the filler loading the higher the water absorption. Similar trend was observed by Shehu *et al.*, (2019).

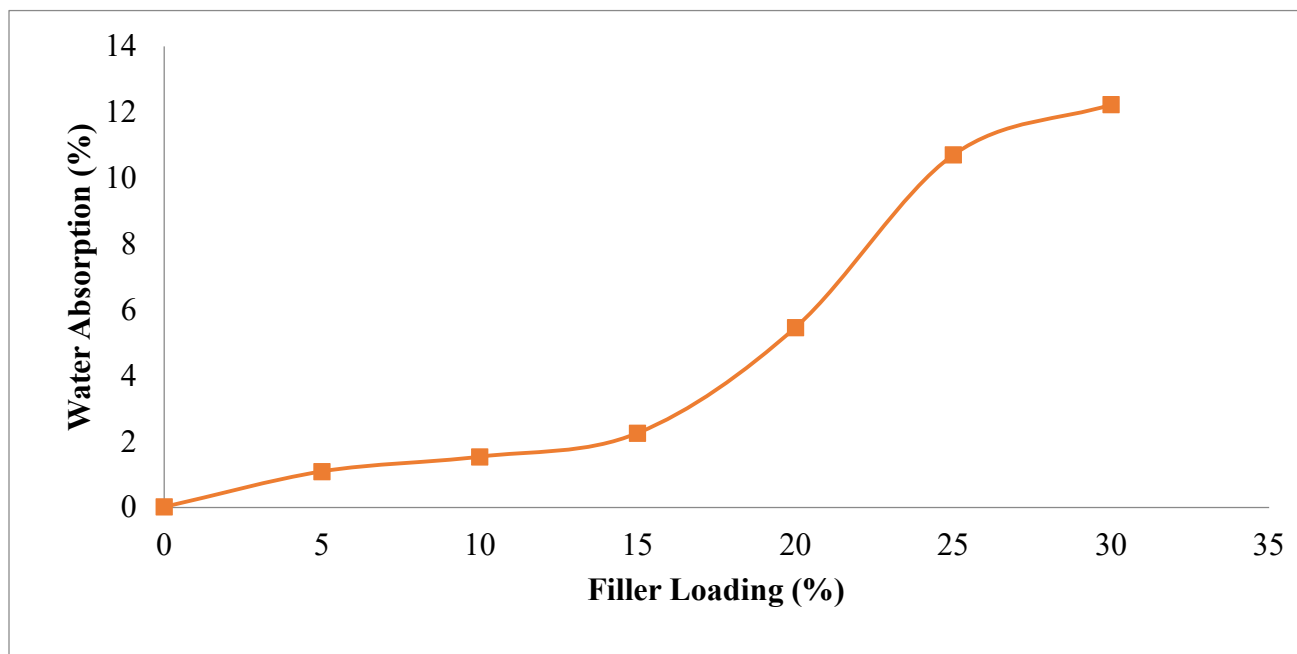


Figure 3.1: Water Absorption versus filler loading (%) for Polyethylene waste / Baobab shell Powder Composites

3.2 Density

The results obtained during this research work on the density of recycled Polyethylene-Baobab shell composites are shown in Figure 4.2. From the graph obtained, it is shown that the density of the composites increased with increase in Baobab shell particles. This behavior was due to the introduction of Baobab shell fillers into the Polyethylene matrix. Incorporation of Baobab shell particles with mass per unit volume of 0.45 gcm^3 into Polyethylene matrix with density of 0.940 gcm^3 which explained why as the filler content increases the density of the composites increased. These findings are in line with the results obtained by Chandrasekar *et al.*, (2023).

The highest density values were seen at 30% with 1.11 gcm^3 while the lowest were observed at 5% with 0.93 gcm^3 . Neat polymer (0%) showed higher density values at 0.94 gcm^3 than 5wt% with 0.93 gcm^3 due to the introduction of fillers into the matrix which may cause air to be trapped in the material causing micro voids in the composites which has adverse effect on the properties of composites hence, reduce the density of the material. The highest density of floor tile composite developed from recycled PET reinforced with natural fiber at 30wt% and 70wt% is 1.688 gcm^3 (Omosebi and Abbas, 2021).

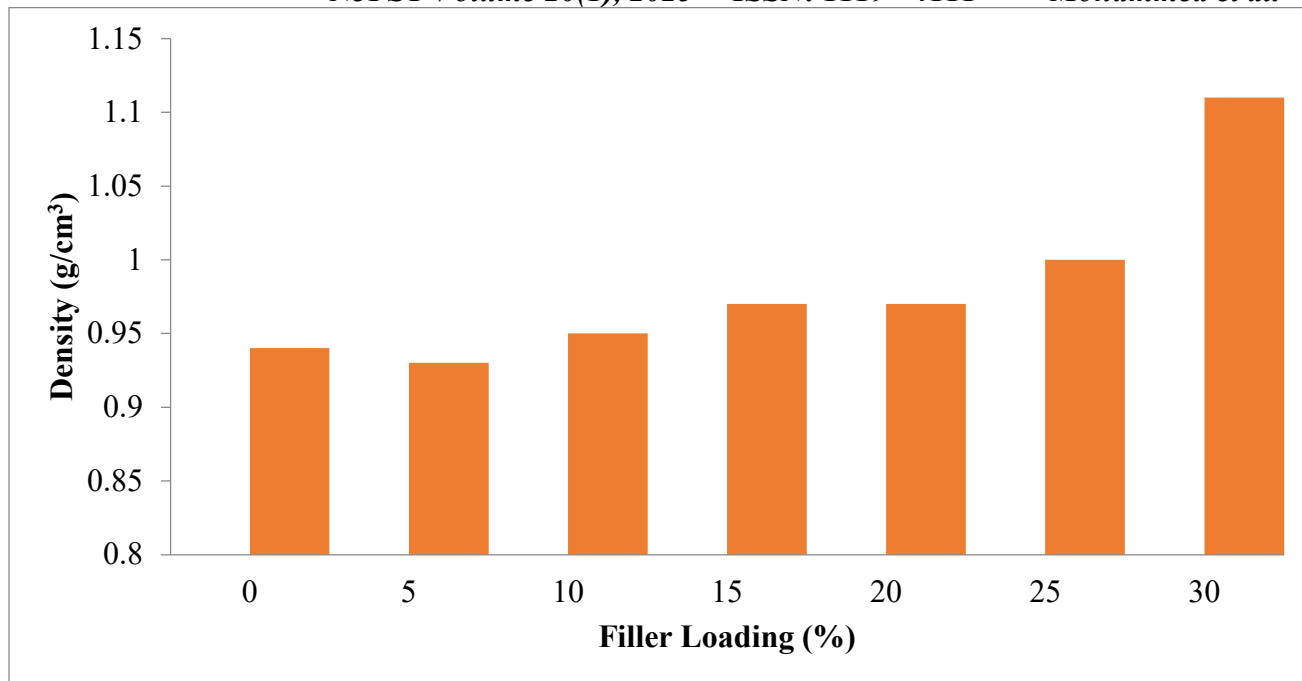


Figure 3.2: Density versus filler loading (%) for waste Polyethylene / Baobab shell Powder Composites

3.3 Tensile Strength

The effects of filler loading on tensile strength of Polyethylene Waste-Baobab shell composites were studied and depicted in Figure 4.1. It was observed that the tensile strength increased from 10.798 MPa to 13.822 MPa for 5% and 15% filler loading respectively. The increase in tensile strength is attributed to better interfacial adhesion between the filler and the matrix as a result of good wettability of the filler particles by the polymer matrix. Maryann *et al.*, (2020) reported that there was an increased tensile strength of the unfilled epoxy (sample A; 0 wt%) from 33 to 48 MPa with the inclusion of 10 wt% fibre into the epoxy matrix. This could be attributed to the ability of the matrix to wet the filler at low concentration. However, at higher filler loading above 15 wt% the tensile strength decreased significantly and remained steady up to 40%. Unfilled polymer (0%) showed lower tensile

strength at 10.722 MPa compared to 5% and 15% filler loading due to ability of the Baobab shell particles to support stress transfer from the matrix. Sudden decreased can be seen with increase in filler loading from 15% to 30% filler loading. This decrease is attributed to weak interfacial adhesion between the Polyethylene Waste and Baobab shell particles as a result of poor wettability. As the filler loading increased the wettability decreased and the matrix begin to lose its binding ability which in turn reduced filler / matrix interfacial interaction therefore caused the tensile strength to decrease. An increase in filler loading leads to the reduction in tensile strength because of uneven distribution of Baobab shell particles in the matrix. This adversely affects the stress transfer from the matrix to the filler. This was also reported by researchers such as Musa *et al.*, (2019), Danladi *et al.*, (2017).

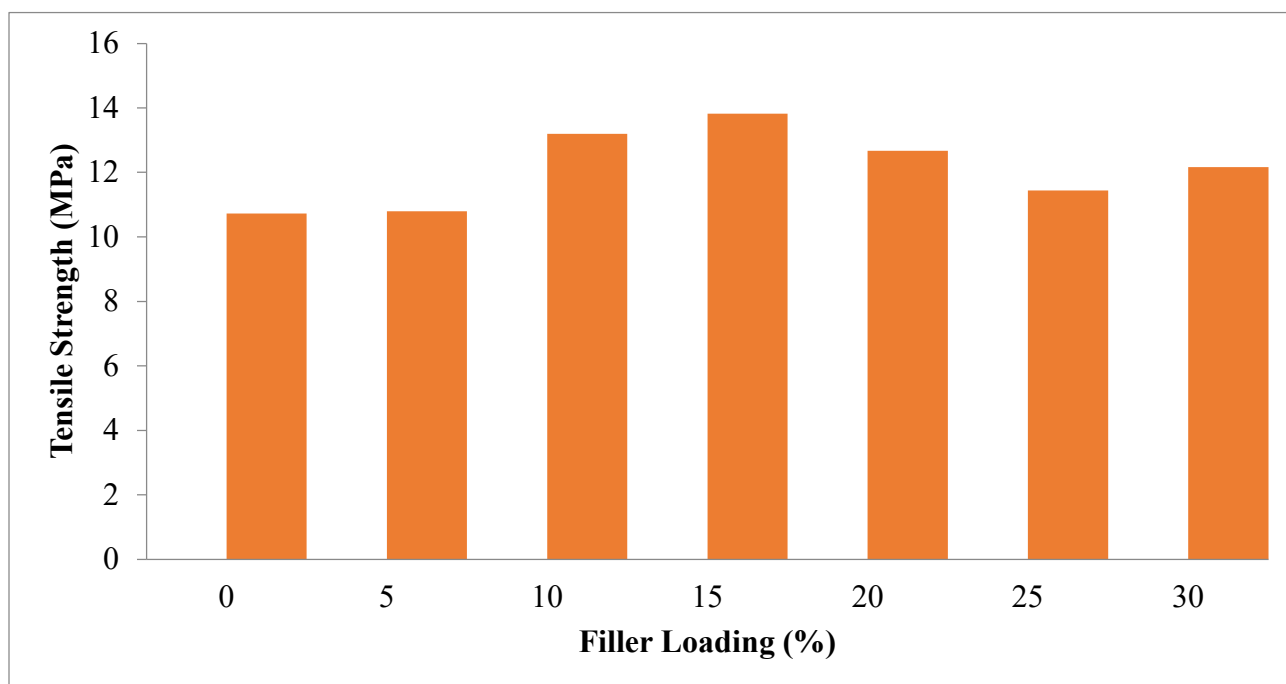


Figure 3.3: Tensile Strength versus filler loading (%) for waste Polyethylene / Baobab shell Powder Composites

3.4 Elongation at Break

The effect of filler loading on elongation at break of Polyethylene Waste-Baobab shell composites are given in Figure 4.2. The elongation at break of the composites increased with increase in filler loading from 9.387% at 0 wt% BFS to 15.153% at 15wt% BFS, thereafter a significant decrease was observed from 10.425% and 8.404% at 25wt% and 30wt% BFS respectively. This observation highlights the fact that the incorporation of much filler into polymer matrix improves the stiffness of the composites thus reducing toughness, that is to say the addition of the rigid filler decreased the ductility of the

polymer matrix. This is because of the reduction in the bond strength at the filler / matrix interface. Similar trend has been reported by other researchers such as Muktari *et al.*, (2019). He found out that incorporation of rigid filler into the polymer matrix decreased the flexibility of low-density polyethylene. From Figure 4.2 it can be seen that 20% filler loading had lower tensile strength of 9.566 MPa than 25% filler loading with 10.425 MPa. This could be due to voids formation during processing. Furthermore, incorporation of more filler into the polymer matrix reduced its molecular chain mobility leading to a decrease in elongation at break.

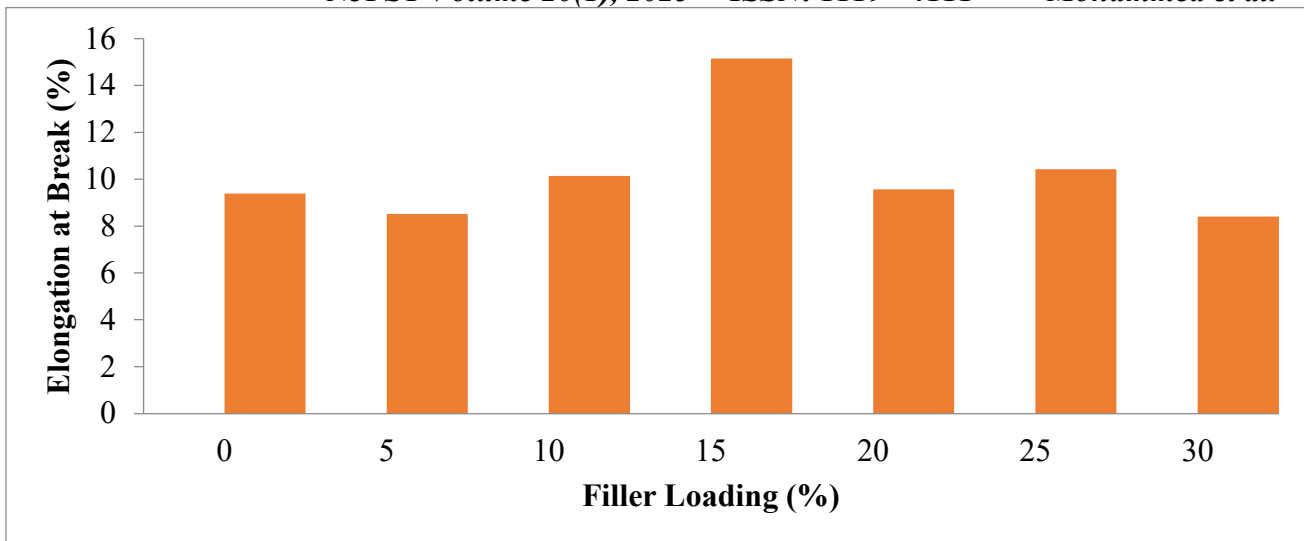


Figure 3.4: Elongation at Break versus filler loading (%) for waste Polyethylene / Baobab shell Powder Composites

3.5 Tensile Modulus

The results of tensile modulus of Polyethylene Waste-Baobab shell composites are presented in Figure 4.3. From the results a progressive increase was observed from 0% to 30% filler loading. This behaviour is due to an increase in stiffness of the material as a result of restriction or reduction in polymer molecular chain mobility which improved the material's rigidity. The incorporation of the Baobab shell particles into the matrix attributed to the stiffening effect and

rigidity to the composites. The lowest tensile modulus was seen at 15% and 25% filler loading with 0.871 GPa and 1.097 GPa respectively. The deviation from this linearity as noticed from decline in 15%, and 25% is believed to have occasioned by initial matrix cracking and the end of the curve represents the ultimate stress which may have been as a result of agglomeration of Baobab shell particles within the Polyethylene matrix. Similar event was observed by Adekomaya and Adama, (2017).

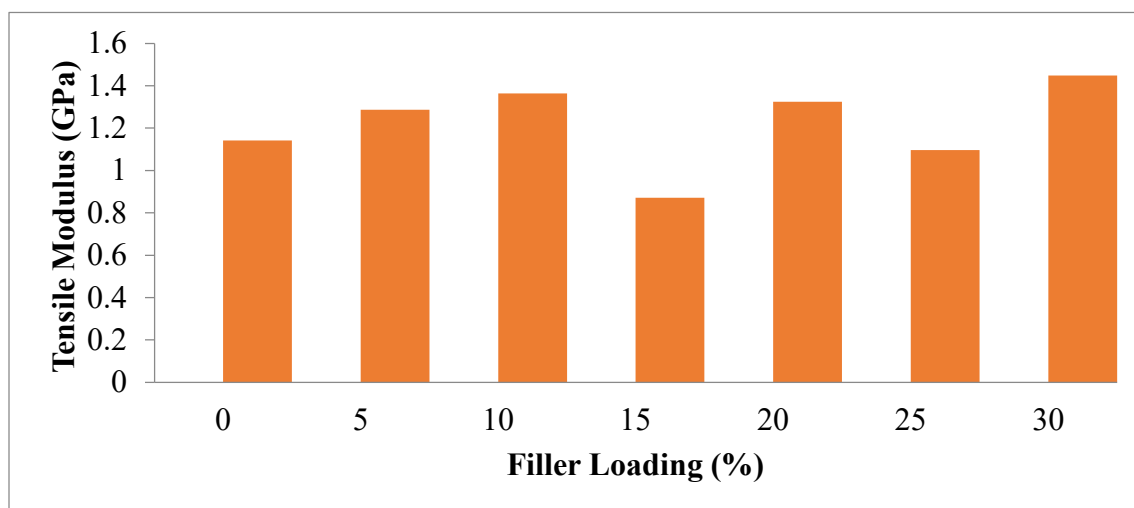


Figure 3.5: Tensile Modulus versus filler loading (%) for waste Polyethylene / Baobab shell Powder Composites

3.6 Morphological Properties

The effect of Polyethylene Waste-Baobab shell particles interaction was investigated using scanning electron microscopy (SEM) PHENOM PRO X (Phenomworld, Eindhoven, Netherlands) as shown in Plates I and II. 15% and 30% BFS samples were considered for microstructural analysis as due the best properties they shown. SEM micrograph of 15% Polyethylene Waste-Baobab shell composites depicted in Plate I showed better dispersion of filler particles within the polymer matrix structure. This is the reason for higher strength of mechanical properties such as hardness and compressive strength. From Plate I few agglomerations and crack was seen due to increase in the filler particles concentration within the polymer matrix, attributed to the low

impact strength. Plate II shows more agglomerated filler particles and voids at 30% BFS loading compared to Plate I at 15% BFS loading. This is due to higher BFS particles concentration incorporated into the matrix leading to poor filler / matrix interaction as a result of difficulties associated with the dispersion of high filler content within the matrix. Wettability is always difficult at higher filler loading leading to poor / weak interfacial bonding between the filler and the matrix which caused the formation of voids during processing. Some authors (Ranganna *et al.* 2012) have reported that the development of voids in any composite is an indication of poor interfacial bond between the fiber and the matrix.

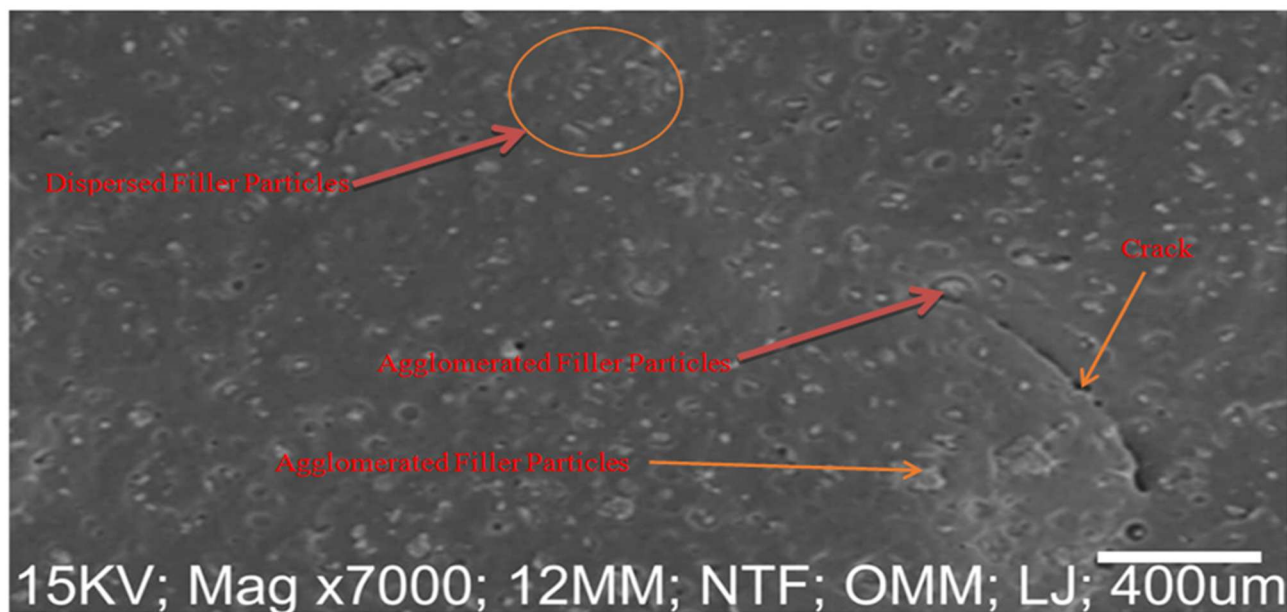


Plate I: SEM Micrograph of 15% Polyethylene Waste - Baobab Shell powder Composite

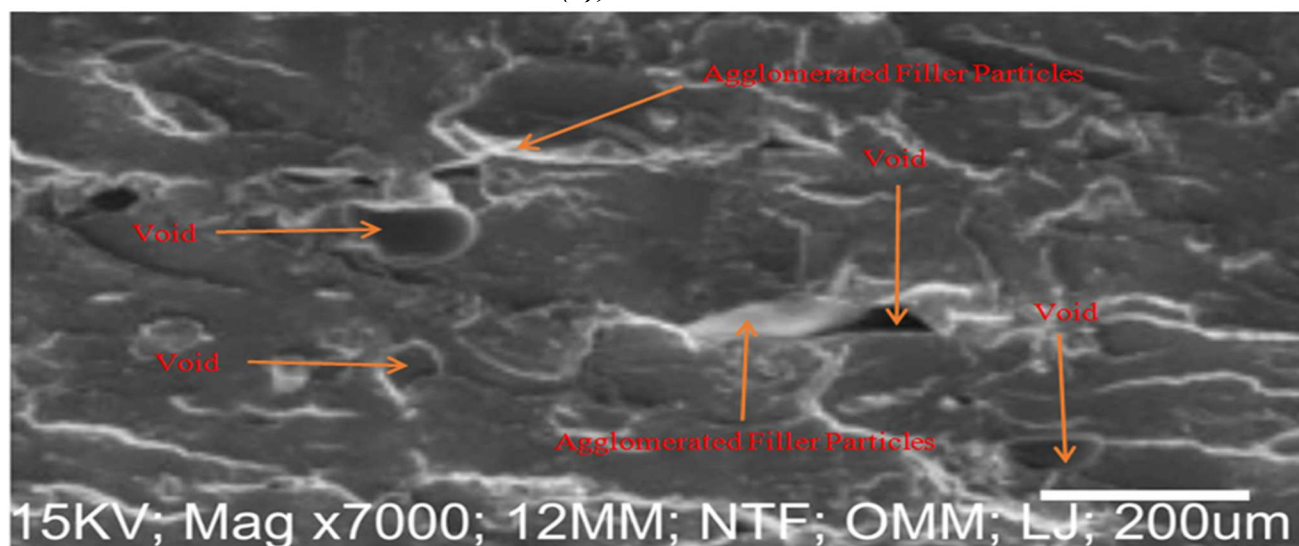


Plate I: SEM Micrograph of 30% Polyethylene Waste - Baobab Shell powder Composite

Conclusion

In this research, baobab powder was mixed with polyethylene waste at seven different ratios for the development of baobab polyethylene waste composites. Physical and tensile properties were investigated using ASTM standards. It was established that the addition of baobab filler in to polyethylene waste increased properties of the composite such as water absorption, density, tensile strength, tensile modulus and a decrease in elongation at break. Water absorption was increased to 12.24 % from 0.02 %. and the density value from 930 kgm^{-3} to 1110 kgm^{-3} . The tensile properties showed an increase of Tensile strength from 10.722 MPa to 13.827 MPa, Tensile modulus from 0.871 GPa to 1.448 GPa. Elongation at break was reduced from 15.153% to 8.404%.

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